Research Article

Invitation to a Public Debate on Ethical Computer Use

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Abstract

To obtain ethically defensible behavior in a particular situation, or context, one must first define what is considered ethically defensible in that context. Without contextual norms, people assume no norms and are then (mis)guided solely by their own experience. According to Integrated Social Contracts Theory, typical contexts include individual-only situations and situations in which the individual is a member of professional, business and social organizations. Norm development is a complex process. It requires discussion between many individuals to identify the ethical issues relevant to each context, to define ethically defensible behaviors for each, and then to decide which norms take priority in the event of a conflict. This paper argues that the importance of context identifies fundamental flaws in the notion that professional associations can address all ethical behaviors through generic codes of ethics. The arguement goes further to assert that Integrated Social Contracts Theory, if applied to universal codes of ethics, can facilitate their redefinition into a useful set of guidelines for ethical professional behavior.

ACM Categories: K 4.1, K 4.2

Keywords: computer ethics, privacy, codes of ethics

Introduction

The incidence of questionable use of computers is increasing in its visibility (Conger, 1997; Pracht, 2000; Sager et al., 2000). We see people at work taking documentation and software. We experience the negative effects of viruses (Conger, 1997; Hafner & Markoff, 1990). Electronic thieves enter computers and steal millions of dollars worth of software, customer files, and competitive information (Pracht, 2000). Internet stalking is increasing, as are other forms of electronic harassment such as bombarding e-mail boxes with advertising or meaningless messages (Bedell, 1999). These are a few of the visible forms of potentially unethical computer use.

There are invisible forms of potentially unethical computer use, too. Companies monitor employee keystrokes, files, and electronic conversations (George, 1994). Information systems personnel

Acknowledgements

We wish to thank Jeff Smith, Janice Sipior, and the anonymous reviewers who helped us develop and refine our thoughts, and who have contributed materially to this work.

observe experts to obtain their knowledge for expert systems development. Employees casually surf company files for tidbits of gossip to tell their friends (Conger, 1997). Companies buy information on consumers from outside sources to match to their customer files but fail to verify the accuracy or validity of such information (Culnan, 1993; Milberg et al., 1995; Smith, 1994; Smith et al., 1996). Information systems (IS) developers build complex systems they cannot fully test, potentially placing bug-ridden code into production. They build systems to support new forms of redesigned business processes, without conscious regard for, or participation from, the people who are the targets of the changes.

Computer ethics is not only a local issue. Internationally, different views on Internet privacy, intellectual property and copyrights, pornography, and personal rights have all led to global tensions and, sometimes, legal remedies (Bresnahan, 1998; EU Directive on Data Protection,1995).

At their core, computer ethical issues are about ownership, responsibility, personal privacy, and access (Mason, 1986; Conger et al., 1995; Loch & Conger, 1995). *Ownership* issues concern the rights to intellectual, physical, and meta-resources. Intellectual property is a new concept for many countries where traditionally artistic creations were not the property of their creator (Velasquez, 2000). In most Asian countries, the highest honor afforded creators is for their work to be copied. While most governments have enacted copyright laws and other legal measures to follow Western ways, weak enforcement and slowly changing local norms present a challenge to the property owners and the local governments. Similar questions arise with regard to computer resources — time, software, hardware, networks, etc.

Responsibility issues deal with who is responsible for completeness and accuracy of work products. Group projects, the most common form of information systems development, frequently suffer from a lack of defined responsibility for product completion and accuracy; consequently, no one accepts responsibility and bug-ridden products are the norm (Conger et al., 1995).

Personal *privacy* issues include collection, usage, access and accuracy of information (Milberg et al., 1995). Collection and usage of personal information revolves around the question of informed consent. For instance, you provide name, address, phone, and credit card information any time you buy on the Internet. You knowingly provide that information for the specific purpose of completing the transaction with the company. Unbeknownst to you, or deceptively communicated (Tedeschi, 2000), the company sells your information to other companies who might then spam you with unwanted mail (Bedell, 1998). The Internet advertising company, Doubleclick.com, tracks surfer activity, combining its online profiles with information from direct mailers and others to determine the actual identity of the Web surfer. On January 27, 2000, a lawsuit was filed against the company alleging violation of privacy rights and deceptive business practices (Tedeschi, 2000). In response to public outcry, Kevin O'Connor, CEO of Doubleclick, publicly announced, "It is clear from these discussions that I made a mistake... in absence of government and industry privacy standards." (Press Release, March 2, 2000). The heart of the issue is whether secondary use of information should be made known to consumers, giving them the right to 'opt out' of such information sales (Culnan, 1993).

Conversely, the European Union has taken an 'opt-in' view of the use of personal information for secondary purposes, enacting the stringent "Directive 95/46/EC of the European Parliament and of the Council of 24 October 1995 on the protection of individuals with regard to the processing of personal data and on the free movement of such data." The directive requires companies to give consumers information about how the data will be stored and used, and for how long, in addition to providing the right to 'opt in.' The directive guarantees European citizens absolute control over their own personal data and prohibits the export of personal data to countries that do not meet EU privacy standards. A truce, known as the "safe harbor" agreement (European Union Safe Harbor Directive, 2000), has the twin goals of ensuring high data protection standards while maintaining the free flow of data cross the Atlantic. A US company that follows EU privacy guidelines will have "safe harbor" from legal action.

Access issues address 'need to know' versus 'right to know' issues (Conger et al., 1995). The

'need to know' issue deals with restricting access to data needed to complete individuals' jobs. As many as 40% of companies in the US do not restrict access to any company files although 60% of companies are concerned about access issues (Anonymous, 1999).

We have always had rules that we internalize and live by so discussions about ethicality and acceptability of behaviors are not new. Although they may ultimately be codified as laws, these rules are the product of a social process that results in defining ethically defensible behavior in a given context. The development of these rules occurs in response to changes in society, which result in new social agreements, ethical codes, and legislation. To exemplify the process, what was once ethical (e.g., slavery in the US) is now unethical. In the computer world, hacking once was viewed as the display of technical prowess. Today, public sentiment supports the legal response to make hacking punishable by a five-year mandatory prison sentence.

Until the 1980s, society's rate of change and the corresponding dialogue took place at what we today might call a snail's pace. Over the last two decades, an accelerating rate of social change has been driven by information technology as it enables new behaviors, so much so that it has become impossible for the evolution of society's rules to keep pace. It is no surprise, then, that the rapid diffusion of the Internet has intensified calls for the development of ethical norms and legislation regarding these issues. One approach has been the development of generic codes of ethics attempting to address the concerns of all parties in all milieus: society at large, nations, companies, professionals organizations, and individuals. The Association for Computing Machinery (ACM) code has one such generic code (Anderson, et al., 1993; Gotterbarn, 1998; Oz, 1993).

While such efforts are commendable, we posit that such generic codes *cannot* effectively address all ethical behaviors because they attempt to span milieus over which they have limited influence. How then might we address the challenge of determining what is ethically defensible technology-enabled behavior across milieus in this rapidly changing world? We propose Integrative Social Contracts Theory (ISCT) (Donaldson, 1982; Donaldson & Dunfee, 1994; Donaldson, 1989)[1] to guide the definition of ethical behavioral norm development and regulation. ISCT proposes consensual, milieu-based development of norms as a basis for developing ethically defensible behaviors and norms.

The discussion of norms begins with the notion of *hypernorms*. Donaldson and Dunfee define hypernorms as those behaviors that are "guided by principles so fundamental to human existence that that they reflect a convergence of religious, philosophical, and cultural beliefs" (1994, p. 265). For example, 'do not kill' and 'do not steal' would be classified as hypernorms accepted globally as standards of behavior.[2] Hypernorms, however, are insufficient to define or guide all behaviors and require elaboration within a hierarchy of milieus. For example, "Western culture, US, public business organizations, ABC Company, people working with computers, people who develop computer applications" might be such a hierarchy. At each level, a *community* with the kinship of a related purpose can be defined.

Through the application of ISCT, it is possible to identify ethical norms within communities (e.g., that computers are for work only), to test these under hypernorms (e.g., do not steal computer resources), and against conflicting norms from other communities (e.g., that computer play can

[1] Donaldson first developed a special social contract theory for business ethics focusing on corporate rights and obligations (1982). Contextually-based social contracts require that the harms be minimized and that trade-offs be consistent with "the general canons of justice" (1982: 53). He broadened the theory in 1989 to include all economic actors and extend the focus beyond a single society to explicitly consider cultural relativism. Integrative social contracts theory (Donaldson & Dunfee, 1994) is based on the recognition of actual norms in different societies and local communities. These local communities specify ethical norms for their respective members through micro-social contracts. These contracts are based on the underlying premise that each member has the freedom to accept, or not, the micro-social contract. The derived norms must be compatible with hypernorms, which is recognition of and agreement on hierarchy. As there will be conflicts among norms, priority is established through the application of rules consistent with the spirit and letter of the resulting macro-social contract.

[2] The term, hypernorm, was derived from an analogy to hypergoods, the term coined by philosopher Charles Taylor (1989) to define goods sufficiently fundamental to serve as a source of evaluation and criticism of community-generated norms. The discussion on hypernorms draws from literature across many disciplines, including anthropology, political science, philosophy, sociology, history, and economics. Hypernorms are also known as global norms and universal norms.

lead to innovation in work), and then to determine which norms should be given priority (Donaldson & Dunfee, 1994).

According to ISCT, the steps to developing a social contract are:

- To determine the appropriate milieu for a given problem,
- To hold discussions among the milieu participants to develop contextual norms,
- To determine how norms relate to those of other milieus,
- To resolve inconsistencies (Donaldson & Dunfee, 1994).

This essay proposes how we, individually and in different milieus, may begin to deal with ethical computing issues by initiating dialogues within the appropriate milieus to develop contextual norms for ethically defensible behavior.

In the next section, we describe how Integrative Social Contracts Theory provides a framework to support the process of norm development within and across milieus. We then apply an existing generic code to several ethical situations to illustrate how the generic code is not an effective means of monitoring computing actions across all milieus. As a better alternative, we suggest a public debate to begin the process of context-specific norm development within and across milieus.

Integrative Social Contracts Theory

We are all citizens of various groups: the global society, a nation, a company, and a professional organization, to name a few. In each of these milieus, we play a role and know the expected behaviors and 'rules' that apply to it. ISCT specifies these milieus as determining the level of the debate.

For instance, the computing issues in this article deal with our roles as individuals in global and national environments, as employees in a company, or as members of a professional group (See Venn diagram in Figure 1). Overlapping sections of the diagram show that one or several possible milieus might apply simultaneously to a given behavior. Within each milieu, some behaviors do not contain ethical computing dilemmas and are of no concern. However, some behaviors do require judgment as to ethically defensible behavior (Mason et al., 1995).

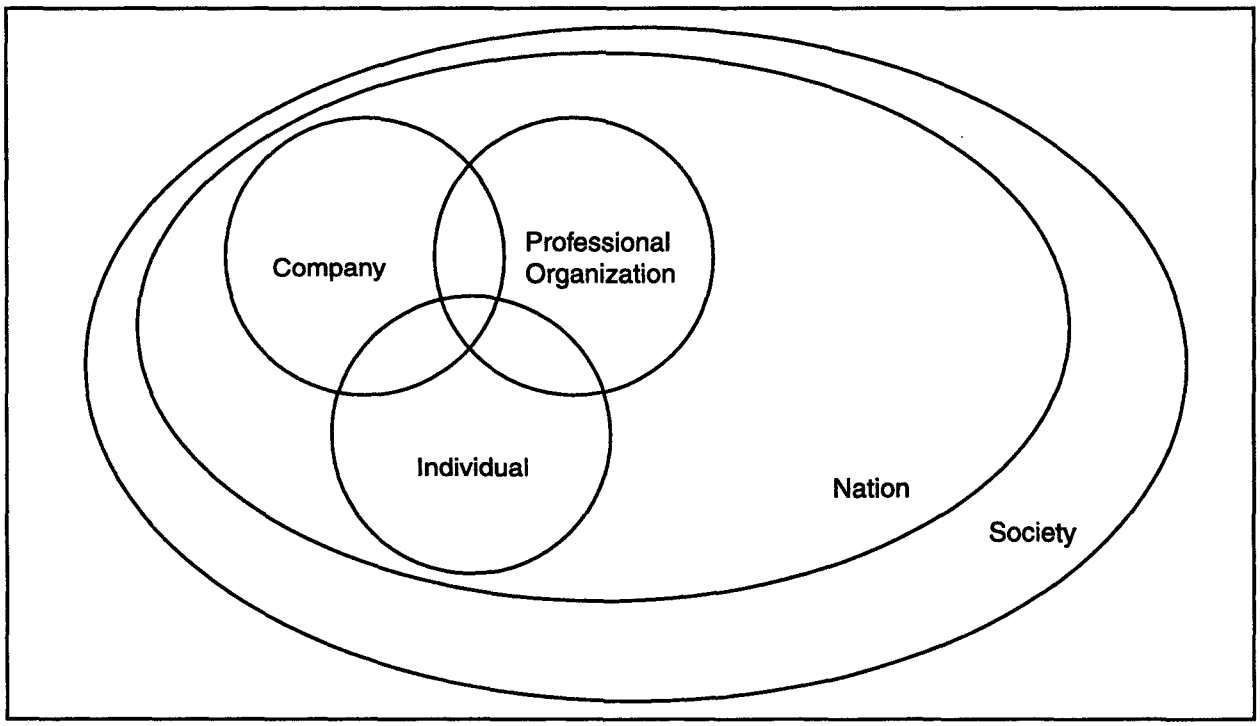


Figure 1. Behavior Milieus

According to ISCT, universal norms guide human behavior in a general sense. Some universal norms, according to philosopher John Locke, might be *do not kill, do not steal,* and *do not abuse children*, (Durant, 1961). Universal norms, however, may not be recognized as applicable to specific actions, or may require redefinition to fit a specific action (Mason et al., 1995). The development of local norms, then, occurs, or should occur, within each milieu for which ethicality is a component of the decisions (Donaldson & Dunfee, 1994).

ISCT posits that a code of ethics in any milieu should not attempt to define all areas of behavior (Donaldson & Dunfee, 1994; Mason et al., 1995). Where there is an absence of defined ethically defensible behaviors and norms, a professional code of ethics, for example, can describe the actions of professionals in performing their specific jobs and may compensate for the lack of a local norm.

Some advocates for professional codes propose that a complete professional code regulate all computer-related behavior (Anderson, et al., 1993; Oz, 1993). However, based on ISCT, this view is misguided. The activities we perform in our professional lives, however many and varied, overlap with only a subset of activities in which we engage in other parts of our lives, as depicted in Figure 1. A professional code, however good, cannot begin to cover the concerns of all individuals, businesses, nations, or society.

For example, the issue of ownership in the information society illustrates how a norm may vary by milieu. At a societal level, the rule, *do not steal*, would apply. A professional code may make a similar generic statement. The individual employee may find however, that what is defined as stealing computer resources in one company is viewed differently by another company. In Microsoft, constructive computer play and personal use of computers are considered *appropriate* use of company computer resources. At IBM, similar activity is considered *inappropriate* use of company computer resources, and in essence, stealing computer and company time that should be used for more constructive purposes (George, 1996; Webster & Martocchio, 1992). Employees at Microsoft and IBM may very well subscribe to the hypernorm of *do not steal* yet the application of that norm, and the associated ethically defensible behavior at the local level, are quite different. This example shows that, even within milieus, micro-level norms develop and vary. Ownership definitions also differ across countries and even differences within the same company across countries may exist (Loch et al., 1998; Velasquez, 2000).

Milieu contention presents another issue — that of determining which issues belong to which milieu, and which issues transverse milieus, thereby requiring a higher-level dialogue. Even with a narrowly defined code of ethics for each milieu, rules in each milieu may overlap and conflict, in which case, individual judgment is still required (Mason et al., 1995; Donaldson & Dunfee, 1994).

The desired and expected end result of milieu norm development is not to remove individual judgment; but rather, to guide our behavior in recognizing *moments of truth* (Mason et al., 1995) requiring ethical reflection and ultimately to support the development of ethically defensible decisions. Codes of ethics that reflect milieu specificity and address cross-milieu contention enrich our perspectives for reasoning through problems.

While we propose the notion of ISCT as it applies to computing ethics, we do not pretend to know which issues belong in which milieu. This discussion is also part of the debate. Four questions that should be discussed in each milieu include:

1. What issues should be in a code of ethics for this context?
2. Who should be the representatives in a debate for this milieu?
3. What is the permeability and rigidity of this milieu?
4. How should this milieu interact with other milieus on each particular issue?

Integrative Social Contracts Theory and Professional Codes of Ethics

In this section, we examine the ACM Code of Ethics, a representative professional code of ethics, in the context of the preceding discussion. Examination of questionable computing situations illustrates the ambiguity of existing professional codes of computing ethics, and the need for local milieu involvement in their development. The current ACM Code of Ethics and Professional

1. General Moral Imperatives. As an ACM member I will...
   - 1.1 Contribute to society and human well-being.
   - 1.2 Avoid harm to others.
   - 1.3 Be honest and trustworthy.
   - 1.4 Be fair and take action not to discriminate.
   - 1.5 Honor property rights including copyrights and patents.
   - 1.6 Give proper credit for intellectual property
   - 1.7 Respect the privacy of others.
   - 1.8 Honor confidentiality.

2. More specific Professional Responsibilities. As an ACM computing professional I will...
   - 2.1 Strive to achieve the highest quality in both the process and products of professional work.
   - 2.2 Acquire and maintain professional competence.
   - 2.3 Know and respect existing laws pertaining to professional work.
   - 2.4 Accept and provide appropriate professional review.
   - 2.5 Give comprehensive and thorough evaluations of computer systems and their impacts, including analysis of possible risks.
   - 2.6 Honor contracts, agreements, and assigned responsibilities.
   - 2.7 Improve public understanding of computing and its consequences.
   - 2.8 Access computing and communication resources only when authorized to do so.

3. Organizational Leadership Impreratives. As an ACM member and an organizational leader, I will...
   - 3.1 Articulate social responsibilities of members of an organizational unit and encourage full acceptance of those responsibilities.
   - 3.2 Manage personnel and resources to design and build information systems that enhance the quality, effectiveness and dignity of working life.
   - 3.3 Acknowledge and support proper and authorized uses of an organization's computing and communication resources.
   - 3.4 Ensure that users and those who will be affected by a computing system have their needs clearly articulated during the assessment and design of requirements; later the system must be validated to meet requirements.
   - 3.5 Articulate and support policies that protect the dignity of users and others affected by a computing system.
   - 3.6 Create opportunities for members of the organization to learn the principles and limitations of computer systems.

4. Compliance with the Code. As an ACM member, I will...
   - 4.1 Uphold and promote the principles of this Code.
   - 4.2 Treat violations of this Code as inconsistent with membership in the ACM.

Table 1. Summary of ACM/ACM Bylaws. Last Update: 06/01/98 by Haritini Kanthou

Conduct was adopted as part of the Bylaws of the organization by the ACM Council on October 16, 1992 (http://www.acm.org/constitution/code.html, last updated, January 16, 1998). The code, as summarized in Table 1, identifies 24 personal responsibility areas to guide information systems professionals' actions in general, professional, and leadership situations.

Professional codes of ethics, such as ACM's, prescribe behavior for privacy, access, ownership, and accuracy. However, the statements are so general as to have little likelihood of affecting behavior. Each issue begs local norms and local accountability of ethically defensible behavior. We argue that professional codes should focus on professional behaviors and that the determination of local norms should be guided by relating a behavior to the applicable, responsible milieu (from the Venn diagram), which then defines its own norms. Underlying our argument for ISCT is the notion that when professional codes attempt

to be all encompassing, their language, impact, and therefore, effectiveness, are diluted. Some examples illustrate this idea.

Company Rights vs. Employee Privacy

A common area of debate focuses on the rights of companies versus rights of employees. In K-Mart and Intel, for instance, electronic surveillance is the norm, while at Apple Computer, it is forbidden (Samuels, 1996). This example illustrates both that different milieus can disagree (e.g. company vs. individual), and that disagreements may occur within a single milieu (e.g. company vs. company).

Milieu contention is also seen by the strikingly different approaches to this issue. Globally, nations set their own policies relating to private industry and, in permissive nations, companies set their own policies on such issues as employee surveillance. The US, as a *permissive* nation, has no formal policy on this issue, so companies, through policies, and focused litigation, have been settling the issue. The company position is that surveillance ensures productivity and enforces security. Companies have been supported in court decisions based on equipment ownership by the employer (George, 1996).

A marked contrast to this approach is unfolding in Europe and other parts of the world. In the European Union, employee rights are paramount and companies may not surveil without full disclosure to their employees (EU Directive on Data Protection, 1995). The effectiveness of a professional code that intersects but does not encompass all these milieus is limited. The ACM Code of Ethics does not directly address this topic and none of its rules is clearly applicable. Thus, discussion of ethically defensible behaviors between and within milieus is also needed.

Computer Viruses

The acceptability of virus development and deployment, which are frowned upon at best, are not specifically addressed by the ACM Code. According to the Code, developing a virus could fall under the general category of *not causing harm*. Yet, Robert Morse, who developed the virus that brought the Internet to its knees, never intended damage (Hafner & Markoff, 1990). Morse operated on Bell Laboratories (now Lucent Technologies) mores, which approved of experimentation in the interest of genuine learning (Hafner & Markoff, 1990). Morse, sadly, had not learned how to test — something not covered by the ACM or Bell Labs codes.

Morse's goal was to see how many computers his program could visit throughout the Internet (Hafner & Markoff, 1990). Some would argue that this is not an ethical issue but a legal one, i.e., breaking and entering. Others would argue that this is not an ethical issue because it is non-destructive. The ACM code is silent on this issue. Arguably, the appropriate milieu for dialogue should be either the national society or global society, since his actions could have affected computers all over the world. His actions did affect 17,000 computers in the US

The Morse example shows just how complex some ethical computing issues can be. No one set of rules applied in this case. Morse was convicted based on the US laws of breaking and entering. He operated according to Bell Labs' norms. Under the ACM Code, his status is unclear. Use of the ACM code suggests using intention in determining responsibility. The law says intention only distinguishes lesser and greater extent of a crime, but there is still a crime.

The WWW and Intellectual Property

The Internet, as a borderless entity, provides opportunities to improve education, commercial activities, and the quality of life. Yet, the Internet has also generated a growing set of unique challenges for developing and applying a code of ethics. One issue is whether all Web sites and their contents are covered by copyright law simply because they are on the Web. Once a venue for open, free sharing of academic computer resources, the Web is now a global, commercial highway. The transition from an open, "anything goes" environment to a commercial environment shows how difficult it is to replace a *de facto* rule of academic sharing with a *de jure* rule resulting from Web commercialization. The ACM Code says:

- Know and respect existing laws pertaining to professional work,
- Avoid harm to others,
- Be honest and trustworthy,

- Honor property rights including copyrights and patents.

Through these prescriptions, the ACM code offers only general moral imperatives and limited professional guidance that might indirectly address Intellectual Property.

On the Internet, norms for behavior should be a global construction, addressed through agencies such as the United Nations, since it represents global user communities. As different societies have different views of ethically defensible behavior and (un)ethical computer uses, the social contracts dialogue becomes paramount. Larger social, ethical issues, such as wealth and knowledge sharing to provide Web access to less developed countries, are other global issues that might be discussed (Conger, 1997; Mason, 1986). The Web has many issues for which the United Nations can fulfill its charter of providing the milieu for negotiations on a global scale.

Other issues with Professional Codes

Professional codes, in trying to be exhaustive, can fail to address some of their own profession's essential activities. The ACM Code, for instance, does not address the use of novice IS developers, untested methods, unclear lines of responsibility and accountability, or the meaning of "full disclosure" on application risks. These issues may or may not be ethical issues, depending on the circumstances of their occurrence.

For instance, the use of untested methods in application development is standard in learning new techniques but increases the risks of failure and bugs for its initial use. In one example, developers believed that their first use of object-oriented methods would allow them to deliver a product sooner than other methods might allow. This goal seems reasonable until one learns that the "end-users" are critically ill and that the application is imbedded in a medical monitoring device. A medical company employee, when asked how the ACM Code of Ethics might apply to this situation, responded, "Does it apply?" To him, there was no risk in their untested use of object-oriented methods because his project "complied with government standards" (Anon., 1994). Compliance with standards was the only concern. The interviewee said, "I believe that the government standard covered the risk and that compliance to [the government standard] alleviated all of my company's risk." The company and its employees never questioned the design methods or the eventual risk to patients hooked to the equipment (Anon., 1994). Honoring contractual obligations overshadowed the possible ethical issues and led to developer abdication of risk responsibility.

In this example, to do good was definitely a goal. Yet, the manufacturer's perspective was simply to meet contractual obligations at the least cost, without acknowledging potential downstream consequences. How does one arrive at the recognition of a moment of truth that

1) an ethical situation — the use of untried methods in developing life-sustaining software — might be present, and
2) the company has more than a contractual obligation; they have an obligation to the "end-user" hospital patient whose life might be in the balance (Mason et al., 1995)?

A code of ethics for computing within the health industry and/or within health-related computing companies should and could address such issues. In this example, ISCT might be applied effectively by the health-care industry. The ACM Code of Ethics, as a generic code, does not, and probably cannot, sufficiently address health care issues. Similar to the problem of addressing permissive and non-permissive organization policies, organizations need guidelines relating to specific entities such as their end-user community (e.g., hospital patients), industry (e.g., nuclear reactor control development), methods (e.g., tested versus untested), or professional skill level (e.g., the assignment of junior versus senior people with industry expertise).

The purpose of the above discussion is not to criticize any one particular code, but rather to demonstrate how even a good professional code does not, and probably cannot, address specific business or societal situations. Issues such as these are governed by companies, nations, or society, and may or may not involve a professional organization.

Alternative Theories

ISCT provides a practical, contextual basis for developing norms in computer ethics. However, other ethical theories may be applied more appro-

priately than ISCT. In this section, we consider deontological and teleological ethical theories as alternatives to ISCT.

Let us consider first a deontological approach that relies on the individual's sense of moral obligation to social *rights* and *duties*. Rights include individual safeguards such as the right to privacy and the right to own property (Mason, et al., 1995). Some duties might include items in the ACM Code such as not harming others through one's computing actions, to be truthful, and to be fair in dealing with others.

Drawing on the last scenario, the medical system developer's actions complied with government standards, honoring his contractual obligations. He followed his *duty* as defined by his company. If "honor contractual agreements" is the only applicable rule, then the medical developer acted ethically. However, if the rule "do no harm" also applies, then the developer faced an ethical moment of truth requiring consideration of whether or not use of untested methods might result in harm to the end 'user' of the application — a patient. In fact, the company failed to articulate a position with respect to untested methods, leading the developer to believe that the ethically defensible behavior was the application of the one known rule — comply with the contract. Narrow application of deontological reasoning might yield this result. Therefore, the question becomes one of moving the discussion beyond the company and its obligations. Norm development that would move the focus beyond the company is a social construction that deontological ethics does not completely address. ISCT provides a means to establish the focus at a more inclusive milieu level and to develop the rules through discussion and debate by vested parties.

Agent-oriented theories, a form of teleological theories, focus on the consequences of the stakeholders' actions. Three types of consequential actions are egoism (acting in self-interest), utilitarianism (acting in the interest of the majority of stakeholders), and altruism (acting to benefit others) (Kalman & Grillo, 1996). Stakeholder theory enumerates all the agents, or stakeholders, in the situation, and then considers the consequences of an action on each stakeholder. Ethically defensible actions are decided based on minimizing harm through analysis of potential outcomes and their likely benefits and costs (i.e., harm).

Drawing again on the last scenario, if the company and the medical system developer are the sole stakeholders, their egoistic interest is in moving product to market in the most efficacious way possible. However, unknown potential failure of critical, life-support equipment does not meet the interests of the stakeholder-patient. Similarly, a stakeholder-doctor may view the company's behavior as placing her in danger of malpractice allegations. If the developer considered the downstream stakeholders, an ethical moment of truth required consideration of whether or not the use of untested methods might result in harm to other stakeholders. It required him to move outside his egoistical sphere of reference and toward either utilitarian or altruistic analysis of the situation.

Recognition that a larger issue beyond the company may exist (and, in this example, does) requires consideration of the issues outside of the immediate context. As with deontological theories, teleological theories do not address the process of norm development, only the application of norms. ISCT provides a framework for the development of norms to guide in identifying milieu contention. This analysis highlights that neither class of ethical theories is inconsistent with ISCT for *defining* norms. In fact, the application of teleological and deontological theories within the framework of ISCT would provide a comprehensive process. Through the process of local communities specifying ethical norms for their members, the context is clarified, the stakeholders are engaged in the dialogue, and the dialogue process permits the exploration of the rights and duties of each stakeholder and the consequences of each possible action.

Objections to Social Contract Theory

To summarize our argument, what is and is not "ethical" is determined by debate within appropriate milieus. Consensus on ethical norms at each level acts as a social contract to guide the behavior of actors within that milieu.

Several objections to this argument seem likely. First, formal debate and complete representation are highly unlikely in many milieus. This argument has some merit in that it is unlikely that all stake-

holders would participate in the discussions. Practically speaking, a code of ethics for a large group, for example a multinational corporation, a country, or even a professional organization such as ACM, will be best developed by a small group, such as the group that developed the ACM code. Consensus via a vote on the proposed code of ethics is a practical solution to the inclusion issue.

A recent example from a different field exemplifies the ISCT process. Medical groups have begun defining ethical issues relating to genome identification, medical information privacy on the web, and other related issues. The proposed norms were developed by a group of representatives and are now out for review and comment in the larger community through professional, business, and industry organizations (Dash, 2000). This example shows how all affected parties may comment and be part of the process even though a smaller group developed the norms.

Another objection is that the social contract resulting from the debate implies total acceptance by all milieu actors. The implied acceptance is highly unlikely and many stakeholders may be unaware of, let alone compliant with, their contractual obligations. This objection applies to any code, developed by any group, for any collective. The purpose of a code is not to gain 100% agreement with it, but to try to gain 100% compliance to collectively defined norms. Norms and their compliance are similar to laws set by government representatives. The rule of the collective dictates ethically defensible behaviors of its members. Like laws, the definition of behavioral guidelines subscribed to by a defining group is ultimately subject to the test of broad acceptance. The application of ISCT front-loads the norm defining process to 1) engage parties, 2) recognize context specificity, and 3) define accepted behaviors within the milieu.

Last, in the absence of norms in other milieus, some might argue that a universal code is better than none. The problem is that a universal code cannot accommodate the range of behaviors accepted in different countries, companies, and groups. The point of this discussion is to argue that each milieu develop its own specific norms and does not presume that, lacking norms, one milieu should define norms for another. Consistent with ISCT is the process of debate across milieus whereby hypernorms are developed.

Summary

Definition of accepted behaviors is critical to obtaining those behaviors. Without the direction provided by local norms, people assume none and behave accordingly. Integrative Social Contracts Theory presents an effective process for developing local norms. Thus, the steps to be taken are to:

1) decide which ethical issues are appropriate for a given milieu,
2) initiate a dialog on ethical issues within each context, and
3) develop norms of appropriate behavior.

ISCT is compatible with both deontological and teleological ethical reasoning; both forms of reasoning can be used to inform norm development described by the ISCT process.

This discussion has demonstrated an effective method for developing norms and has argued that generic norms are not successful in guiding behavior. Readers are encouraged to reflect on the ethical issues, discuss computing ethics in respective workplaces, on the Internet, and in other milieus to which they relate, and to participate in the development of local norms for the contexts in which they live. Readers are further encouraged to take action in any organization with a universal code of ethics to redevelop its code of ethics.

To learn what others are doing in this area, codes defined for different milieus for others to use might be made available on the Internet through a clearinghouse for companies, organizations, and groups. A well-known group such as ACM SIGMIS, or an established ethics center, is encouraged to host such a web site to provide a forum for this dialogue.

References

Anderson, R.E., Johnson, D.G., Gotterbarn, D. , and Perrolle, J. (1993). "Using the ACM Code of Ethics in Decision Making," *Communications of the ACM*, Vol. 36, No. 2, pp. 98-107.

Anonymous. (1994). Private e-mail conversation with the author, April 12, May 8, May 22.

Anonymous. (1999). *CIO Magazine Study Reveals the Top Five Concerns about Online Purchasing.* http://www.cio.com/info/releases/122099_onprchsng.html.

Bedell, D. (1999). "Smashing Spam: Internet Coalition Pushing for Federal Law to Stop Electronic Quick-Buck Artists," *Dallas Morning News*, July 13, pp. 1F.

Bresnahan, J. (1998). "The Business Case for Privacy," *CIO Magazine*, March 15, http://www.cio.com/archive/enteprise/031598_privacy_content.html.

Conger, S.A. (1997). "Ethical and Legal Issues on the Web," *Presentation at the National Informs Meeting*, October 28, Dallas, TX.

Conger, S.A., Loch, K.D., and Helft, B.L. (1995). "Ethics and Information Technology Use: A Factor Analysis of Attitudes to Computer Use," *Information Systems Journal*, Vol 6, No. 4, pp. 161-184.

Culnan, M. (1993). "How Did You Get My Name?: An Exploratory Investigation of Consumer Attitudes Toward Secondary Information Use," *MIS Quarterly*, Vol. 17, No. 3, pp. 341-363.

Dash, J (2000). "Privacy Fears Spur Health Groups' Moves," *Computerworld*, Vol. 34, No. 9,February 28, pp. 8.

Donaldson, T. (1982). *Corporations and Morality*, Englewood Cliffs, NJ: Prentice-Hall.

Donaldson, T. (1989). *The Ethics of International Business*, New York: Oxford University Press.

Donaldson, T., and Dunfee, T.,W. (1994). "Toward a Unified Conception of Business Ethics: Integrative Social Contracts Theory," *Academy of Management Review*, Vol. 19, No. 2, pp. 252-284.

Durant, W. (1961). *The History of Philosophy*, New York: Free Press.

EU Directive on Data Protection (1995). "Directive 95/46/EC of the European Parliament and of the Council of 24 October 1995 on the protection of individuals with regard to the processing of personal data and on the free movement of such data," http://www2.echo.lu/legal/en/dataprot/directiv/directiv.html.

European Union Safe Harbor Directive (2000). http://europa.eu.int/comm/internal_market/en/media/dataprot/news/harbor4.htm.

George, J.F. (1996). "Computer-based Monitoring: Common Perceptions and Empirical Results," *MIS Quarterly*, Vol. 20, No. 4, pp. 459-480.

Gotterbarn, D. (1998). "Raising the Bar: A Software Engineering Code of Ethics and Professional Practice," *Proceedings of the Ethics and Social Impact Component on Shaping Policy in the Information Age, ACM Policy Conference*, Washington D.C., May 10, pp. 26-28.

Hafner, K., and, Markoff, J. (1990). *Cyberpunk: Outlaws and Hackers on the Computer Frontier*. New York: Touchstone.

Kalman, E., and Grillo, J. (1996). *Ethical Decision Making and Information Technology*, 2nd Edition, New York: McGraw-Hill.

Loch, K.D., and Conger, S.A. (1995). "Evaluating Ethical Decision Making in Computer Use," *Communications of the ACM*, Vol. 39, No. 7, pp. 74-83.

Loch, K.D., Conger, S.A., and Oz, E. (1998). "Ownership, Privacy and Monitoring in the Workplace: A Debate on Technology and Ethics," *Journal of Business Ethics*, Vol. 18, No. 2, pp. 197-207.

Mason, R.O. (1986). "Four Ethical Issues for the 1990s," *MIS Quarterly*, Vol. 10, No. 1, pp. 5-14.

Mason, R.O., Mason, F., and Culnan, M.J. (1995). *Information Ethics*, Thousand Oaks, CA: Sage Publications.

Milberg, S.J., Burke, S.J., Smith, H.J., and Kallman, E.A. (1995). "Values, Personal Information Privacy, and Regulatory Approaches," *Communications of the ACM*, Vol. 38, No. 12, pp. 65-74.

Oz, E. (1993). "Ethical Standards for Business Professionals: A Comparative Analysis of Four Major Codes," *Journal of Business Ethics*, Vol. 12, No. 9, pp. 709-726.

Pillar, C. (1993). "Special Report on Electronic Privacy: Bosses with X-ray Eyes," *MacWorld*, July, pp. 118-120.

Pracht, M. (2000). "Internet Terrorism is Expected to Grow," *Global Fax* (January 28) pp. 3.

Sager, I., Hamm, S., Gross, H., Carey, J., and Hof, R. (2000). "CyberCrime," *Business Week*, February 21, pp. 36-42.

Samuels, P.D. (1996). "Who's Reading Your E-mail? Maybe the Boss," *The New York Times*, May 12, pp. 11.

Smith, H.J. (1994). *Managing Privacy: Information Technology and Corporate America,* Chapel Hill, NC: UNC Press.

Smith, H. Jeff, Milberg, S., and Burke, S. (1996). "Information Privacy: Measuring Individuals' Concerns About Corporate Practices," *MIS Quarterly*, June.

Taylor, C. (1989). *Sources of the Self*, Cambridge, MA: Harvard University Press.

Tedeschi, B. (2000). "In a Shift, DoubleClick Puts Off Its Plan for Wider Use of Personal Data of the Internet Consumer," *New York Times*, March 3, www.nytimes.com.

Webster, J., and Martocchio, J.J. (1992). "Microcomputer Playfulness: Development of a Measure with Workplace Implications," *MIS Quarterly*, Vol. 16, No. 2, pp. 201-226.

Velasquez, M. (2000). "Globalization and the Failure of Ethics," *Business Ethics Quarterly*, Vol. 10, No. 1, pp. 343-353.

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